NPLify · P0 Technical Baseline

Calculation Specification

New XP Technologies Limited

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How to read this pack

Four map pages are now part of the pack as documents of their own, and are the place to start: the Top-Down Map (from the business, through the data model and the deal states, to the scenarios and the reading order), the Deal Flow on one page (Quote, Collect, Convert, Settle, with the roles and the money path), the Workflow Tree (8 classes, 27 sub-classes, 92 procedures, each naming who initiates, who approves, the codes it is written under and the records it touches) and the Workflow Tree on one page. Read them in that order, then this document. Every worked transaction (T), acceptance vector (L, B), walkthrough (W, X), flow (F) and procedure (C, A, D) has one identifier, listed in the Reading Companion's identifier register; the Test Strategy & Coverage document says which test proves each procedure.

1 · Conventions

- **Rates** are stored normalised to the pair's `quote_direction` (out units per 1 in unit unless the pair says in_per_out) at `CURRENCY_PAIR.rate_precision`. A partner's rate is stored as quoted (`PARTNER_RATE_VERSION.quoted_rate`, `quoted_direction`) and the normalised `rate` is derived: equal to the quote when the directions agree, 1 ÷ quote at the pair's precision when they differ.
- **Amounts** are exact decimals; fiat and USDT amounts are computed to 2 decimals (BTC-like assets to 8) with round half up, then rounded as the pair's option says (§9). Every intermediate figure in this document is computed, not typed.
- **Percentages** are stored as fractions (1.4 % = 0.014). `sender_pays_pct` + `receiver_pays_pct` = `FEE_STRUCTURE.pct`.
- **Precision of a check**: an acceptance vector matches when the computed figure and NPL's figure agree to the cent, or to the unit where NPL's sheet rounds to the unit.

2 · Rate basis, derived rates and the breach check

`FEE_STRUCTURE.rate_basis` says which rate the sender is priced on: **market** (the fee is disclosed; the partner's cost comes out of NPL's fee parts — Evo, the reseller projects) or **agent_rate** (a calculated rate, breakdown not disclosed; NPL keeps the whole fee and the partner's cost sits on top — BF via Ali). The partner's own pricing is `PARTNER_PAIR.partner_pricing`: **fee_in_rate** (its cost is inside its rate, never posted; Ali, assumed 1 %) or **fee_on_market** (it converts at market and states `partner_fee_pct` separately; Jeton 1.25 %, Aquanow 0.80 %).

```
basis_rate = market_rate if rate_basis = market
 = partner_rate_version.rate if rate_basis = agent_rate
rate = quoted_rate if quoted_direction = pair.quote_direction
 = round(1 ÷ quoted_rate, rate_precision) otherwise
spread = 1 − partner_rate ÷ market_rate (what the partner keeps against market, fee_in_rate)
breach = spread > avg_2w_spread × (1 + variance_threshold_rel) → the pair freezes until Finance
decides
```

2a · When the breach check cannot be trusted

The breach check of Section 2 compares today's spread with the average spread of the previous two weeks. Three situations make that comparison unreliable; each has its own rule. The three values the rules use are project thresholds set by NPL Finance (THRESHOLD kinds `rate_min_observations`, `rate_fallback_spread_abs` and `rate_abs_tolerance`); the figures below are illustrative. Stale rates, the freeze on a breached pair and the use of confirmed versions only are unchanged.

Situation	Rule	What the system does
Too little history	With fewer than rate_min_observations confirmed rate versions in the two-week window (a new partner, a new pair, a long holiday), the two-week average is not used.	The spread is compared with the Management-approved fallback threshold instead: breach = spread > rate_fallback_spread_abs. Finance confirms the version with a reason; the pair is not frozen merely for lack of history.
Unusable baseline	When avg_2w_spread is zero or negative (the partner has been pricing at or better than market), the percentage formula has no meaning.	The percentage formula is not run. The fallback threshold applies as above, and the version is flagged for Finance to confirm (the guard of the next row applies: the version is not usable until confirmed).
Unusual rate, favourable or not	A spread outside an absolute tolerance around the two-week average, in either direction: spread < avg_2w_spread − rate_abs_tolerance or spread > avg_2w_spread + rate_abs_tolerance.	The version is flagged, not rejected: it enters the state *flagged* and cannot price a quote or instruct a conversion until Finance confirms it with a reason, or rejects it. The pair itself is not frozen: the last usable version keeps serving until it expires, and a corrected version may be entered. An unusually favourable rate is checked (direction slip, wrong pair, stale partner) before it is used, not after. The existing breach rule still applies on top for an unfavourable spread.

```
observations = count of confirmed PARTNER_RATE_VERSION rows for the partner × pair in the last
14 days
if observations < rate_min_observations or avg_2w_spread ≤ 0:
 breach = spread > rate_fallback_spread_abs (fallback; no percentage formula)
else:
 breach = spread > avg_2w_spread × (1 + variance_threshold_rel) (Section 2, unchanged)
flag_for_verification = |spread − avg_2w_spread| > rate_abs_tolerance (either direction; the
version waits as flagged, unusable; the pair does not freeze)
```

Typical Ali spread 1.00 %, variance_threshold_rel 20 %: a spread above 1.20 % breaches as today. New partner with four confirmed versions and rate_min_observations 10: the average is ignored and a spread above the fallback 1.50 % breaches. A day on which Ali quotes 0.30 % below market (spread −0.30 %) with rate_abs_tolerance 0.50 %: no breach, but the version is flagged and no quote or conversion can use it until Finance has confirmed the direction and the pair; a quote attempted meanwhile is refused and the refusal is logged. The four rules are tested in Phase 1 (M1) with one case each (the fourth: a quote and a conversion attempted against a flagged version, Test Strategy V-RB4).

A flagged version is a guard, not a warning. The states of a version are those of the Lifecycle & Approvals document (Figure 3) with one state added: a new version becomes usable, breached (the pair frozen) or *flagged*; a flagged version becomes usable when Finance confirms it with a reason (confirmed_by, approval_reason), or is rejected, and one that expires unconfirmed is simply expired. Only a usable version prices a quote or instructs a conversion: the engine refuses a quote and a conversion that would use a flagged version, tells the user that the version awaits Finance, and writes the refusal to AUDIT_LOG (record PARTNER_RATE_VERSION, action quote_refused or conversion_refused, with the user and the version). Operations cannot issue the quote first and have Finance check afterwards; there is no monitored-use policy. The acceptance test attempts a quote and a conversion while the flag awaits review: both are refused and the audit record exists; after Finance confirms, the same quote prices on the confirmed version.

3 · Fee parts on the principal

The sender's part and the receiver's part are each a percentage of the **principal at the basis rate**. They are never compounded on each other (300,000 USDT × 0.9000 − 2 % = 264,600).

```
principal = amount_in × basis_rate (fixed_in)
fee_sender_part = principal × sender_pays_pct
fee_receiver_part = principal × receiver_pays_pct
sender_rate = basis_rate × (1 − sender_pays_pct) (the rate the sender is given)
gross_out = amount_in × sender_rate = principal − fee_sender_part
client_net = gross_out − fee_receiver_part = principal × (1 − sender_pays_pct −
receiver_pays_pct)
```

- 10,000 USDT × market 0.9000 = 9,000.00 EUR (the principal at the basis rate)
- sender part = 9,000 × 1.40 % = 126.00; receiver part = 9,000 × 0.60 % = 54.00
- sender rate = 0.9000 × (1 − 1.40 %) = 0.88740; gross out = 10,000 × 0.88740 = 8,874.00
- client net = 9,000 × (1 − 2.00 %) = 8,820.00 = gross out − receiver part

Confirmed rule (NPL, 11 October). Both parts are on the principal. NPL's fee sheet for Evo prints the receiver side as "EUR 25,704 − 0.6 % = 25,550", which takes the receiver's part from the gross out; on the principal it is 25,547.88, a difference of 0.6 % × 1.4 % of the principal (about 0.008 %). The sheet's receiver line is to be brought in line with the rule; the engine follows the rule.

4 · Fixed and variable fee modes; the minimum-margin check

In **fixed** mode `pct` is applied as above. In **variable** mode the fee is whatever leaves NPL its minimum margin after the partner's cost, within a floor and a cap: this is the check that decides whether 2 % is enough when the agent rate is high.

```
required_pct = min_margin_pct + spread if rate_basis = market (the partner's cost comes out of
the fee)
 = min_margin_pct if rate_basis = agent_rate
fee_pct = min(max_markup_pct, max(min_markup_pct, required_pct))
below_margin = fee_pct < required_pct → warn, never refuse a quote; Finance may ask the partner
for a better rate or
 recommend a fee above the cap (raise_fee_cap, Management approves)
```

4a · Who may send a quote, by margin

The rules of Section 4 and of the Lifecycle & Approvals document (section 5, price_quote) in one table.

Situation at pricing	Test	Who sends or approves
Margin at or above the floor	fee_pct ≥ required_pct	Operations sends the quote; no approval.
Below the floor	fee_pct < required_pct (below_margin): the warning becomes a blocker at the send step	Finance approves before the package is sent; Finance may instead ask the partner for a better rate.
Negative margin	fee_pct after the cap leaves earnings < 0 (required_pct > max_markup_pct + the partner's cost)	Finance approves and Management is added (escalation); the quote is not sent without both.
Fee above the cap	the fee needed exceeds max_markup_pct	Management approves a raise_fee_cap after Finance recommends it (FEE_DECISION); otherwise the cap stands.

5 · Fee tiers by monthly volume

```
tier = the FEE_TIER of the structure with the highest min_monthly_volume ≤ PROJECT_MONTH_VOLUME
at quote
portion_1 = max(0, threshold_of_next_tier − PROJECT_MONTH_VOLUME) capped at amount_in, priced
at tier.pct
portion_2 = amount_in − portion_1 priced at next_tier.pct
fee_parts = Σ portions × basis_rate × pct (sender and receiver parts in the structure's
proportion) — the blended sum
```

GDC: 2.5 % up to USD 500,000 a month, 2.25 % above; a leg that takes the month past the threshold is split into two DEAL_FEE_TIER portions.

5a · The month-volume counter and what a void does to it

```
PROJECT_MONTH_VOLUME(project, month, ccy) = opening_volume(project, month, ccy)
 + Σ amount_in × basis_rate of forward legs priced in the calendar month from the opening date,
 in FEE_TIER.volume_currency, excluding legs voided before or after money moved
opening_volume = the approved OPENING_ENTRY of kind month_volume for the project, the opening
month and the currency
 (Cutover Approach §4: the month-to-date Σ amount_in × basis_rate of the deals priced before
the opening date),
 less the volume of any of those deals voided after the opening date; zero in every later month
tier at quote = the tier with the highest min_monthly_volume ≤ PROJECT_MONTH_VOLUME at the
moment of pricing
```

A void lowers the counter for every quote priced after it; it never re-prices a leg already priced (pricing fields are write-once), so a leg that was split across two tiers keeps its split even if an earlier deal in the month is later voided. A return leg and a balance conversion do not count. Two quotes priced at the same moment read the counter under the lock on the month-volume row, so they are ordered and the second sees the first (the one-line database rule of Phase 1, M0 and M3); this is tested with real GDC data rather than designed for every edge case now. A backdated correction is a new leg priced today and counts in today's month.

The opening volume. A project that opens in the middle of a month carries its month-to-date volume into the counter as the opening line (OPENING_ENTRY of kind month_volume: one figure per project, month and volume currency, taken from the sheet, verified by Finance against the deals it sums and approved by Management like every other opening entry; it is a counter, not a posting). Tier evaluation reads the counter and never asks where its volume came from: with GDC at USD 480,000.00 on an

opening day 20, the next target_out of USD 40,000.00 is split at the USD 500,000.00 threshold into 20,000.00 at 2.5 % and 20,000.00 at 2.25 % (L14), and the counter stands at 520,950.00 afterwards. Three rules keep the opening line honest. (1) The system's own legs count only from the opening date: an in-flight deal entered in its current state keeps its original priced_at, before the opening date, so its volume is inside the opening line and is never added a second time. (2) A void after the opening date of a deal whose volume is in the opening line lowers the counter by that deal's volume, exactly as a void of a system-priced leg does; the opening line itself is never edited, the void is what the counter reads. (3) In turnover reporting the opening month shows the opening line as one figure, volume before the opening date from the sheet, beside the system's legs priced from the opening date; the in-flight deals are reported under that line and not as system turnover, and from the following month the opening line is in no figure at all. A project opened on the first of a month has an opening volume of zero.

6 · Amount modes: fixed_in and target_out

A collect-first deal prices what arrived (**fixed_in**, §3). A quote-first deal for a given out amount (**target_out**: an invoice, a GDC USD amount, a reseller's EUR) applies the fee as a markup on the target, which is how NPL's sheets print it (EUR 50,000 × 1.01 ÷ rate; USD 84,482 + 2.5 %).

```
amount_in = target_out × (1 + sender_pays_pct) ÷ basis_rate (then cash-rounded if collected in
cash, §9)
sender_rate = basis_rate ÷ (1 + sender_pays_pct) (printed as target ÷ amount_in)
client_net = target_out × (1 − receiver_pays_pct) + cash_rounding_surplus
fee_sender_part = amount_in × basis_rate − (target_out + cash_rounding_surplus)
```

Note: × (1 + s) and ÷ (1 − s) differ by s^2 (0.01 % at a 1 % fee). NPL's convention, × (1 + s), is adopted; the fixed_in formula keeps × (1 − s). The cash-rounding surplus is the client's (in NPL-GR, NPL-GR's).

- NPL-GR: invoice 9,450 EUR ÷ 0.8708 = 10,852.09 USDT to deliver
- amount_in (exact) = 10,852.09 × (1 + 2 %) × 32.50 THB per USDT = 359,746.78 THB; collected as cash, rounded up to the 1,000 = 360,000 THB
- client net = 360,000 ÷ 32.50 × (1 − 2 %)… no: in target mode the fee is on the target, so client net = target + the cash-rounding surplus = 10,859.73 USDT (7.63 surplus); sender part = amount_in × market − client net = 217.19
- Ali converts at 1 ÷ 32.83: delivered 10,966.15; NPL earnings = 10,966.15 − 10,859.73 = 106.42 = the 2 % markup less Ali's 1 %

7 · Customer-facing figures

```
customer_facing_fee_pct = sender_pays_pct + receiver_pays_pct if customer_facing_fee_basis =
actual
 = sender_pays_pct on the agent rate if sender_pays_on_agent_rate
 = FEE_STRUCTURE.customer_facing_fee_pct if fixed (BF via Ali prints 1 %)
customer_facing_rate = sender_rate ÷ (1 − customer_facing_fee_pct) — the "market rate" the
quote prints, derived, never typed
```

8 · What the partner delivers, and NPL's earnings

```
fee_in_rate: delivered = amount_in × partner_rate; partner_cost (inferred) = amount_in ×
(market − partner_rate), information only
fee_on_market: converted = amount_in × market_rate; partner_cost = converted × partner_fee_pct;
delivered = converted − partner_cost
earnings = delivered − client_net (NPL's margin: EARN_GROSS − EXP_PARTNER)
EARN_GROSS = converted − client_net (fee_on_market: the full fee parts, the partner's cost
```

```
booked apart as EXP_PARTNER;
 fee_in_rate: the fee parts less the partner's hidden cost — equal to earnings)
retained_in = earnings_in_out_currency ÷ partner_rate when the margin is kept in the collection
currency (CONVERSION.residual_in)
converted_in = amount_in − retained_in − offset_in (what the partner actually converts, §13)
```

- Ali (fee_in_rate) converts at 0.8910: delivered = 10,000 × 0.8910 = 8,910.00; NPL earnings = 8,910.00 − 8,820.00 = 90.00; Ali's inferred cost = 10,000 × (0.9000 − 0.8910) = 90.00, information only
- Jeton (fee_on_market 1.25 %) converts at market: 9,000.00; partner cost = 9,000.00 × 1.25 % = 112.50; delivered 8,887.50; fee charged 180.00 (EARN_GROSS); NPL earnings = 8,887.50 − 8,820.00 = 67.50

9 · Rounding and precision

```
payout: amount_rounding = exact | truncate_unit | nearest_unit | round_up_unit | nearest_n |
round_up_n (rounding_unit)
 payable = round(client_net, option); residual = client_net − payable (signed)
 residual > 0 → stays client money in DUE; residual < 0 → NPL's cost, EXP_ROUNDING, out of its
margin
cash in: cash_rounding = truncate_n | nearest_n | round_up_n (cash_rounding_unit: AED 100, USD
cash 10, THB 1,000) — the collection side;
 a cash payout rounds by the pair's amount_rounding (round_up_n 1,000 for THB hand-overs)
 amount_in = round(exact amount_in, option); surplus = (amount_in − exact) × basis_rate × (1 −
fee) → the client's
rates: rate_precision per pair (6 for EUR/USDT, 8 for BTC); intermediate amounts 2 dp (8 for
BTC), round half up
```

Bank transfers go out exact, cents included; rounding to a unit applies where the pair says so, in practice to cash hand-overs (NPL, 11 October). USDT paid by reseller customers is rounded up.

- THB payout in cash: client net 95,844.00 THB, cash_rounding round_up_n 1,000 → 96,000.00; residual = 95,844.00 − 96,000.00 = -156.00 (negative: NPL's cost, EXP_ROUNDING); NPL's pool gets 966.00 -156.00 = 810.00

10 · Same-currency pairs

```
basis_rate = partner_rate = 1; fee parts and client_net as in §3; earnings = fee_sender_part +
fee_receiver_part
```

Evo EUR → EUR: 2.8 %, sender 1.4 % and receiver 1.4 %: 5,000 EUR → client net 4,860.00, NPL 140.00.

11 · Markup shares and rebates

```
share base (share_of_net_markup) = fee_sender_part + fee_receiver_part − partner_cost (rebates
never in the base)
share_i = base × pct_i; Σ pct_i ≤ 100 %; accrued and posted with the conversion
 npl_pays: SHARE_PAYABLE to the party, off EARN_GROSS
 party_retains: the party holds the funds; NPL's remainder = base − Σ shares → SHARE_RECEIVABLE
share_of_earnings: share = earnings × pct; fixed_pct: priced into the sender rate
rebate = amount_in × pct (basis pct_of_amount, the only basis; Jeton 0.4 %, Aquanow 0.3 %, Novi
0.3 %), in PARTNER_REBATE_RULE.payout_currency
 accrued with the conversion → REBATE_RECEIVABLE / EARN_REBATE; the statement clears it, a
difference trues up EARN_REBATE
```

- Raeen sub-account at Aquanow: 10,000 × 0.9000 = 9,000; sender pays 4 % = 360.00; client net 8,640.00; Aquanow 0.80 % of 9,000 = 72.00; delivered 8,928.00

- share base = fee parts 360.00 − partner cost 72.00 = 288.00; Raeen 50 % = 144.00; NPL remainder 144.00; the 0.3 % rebate (30.00 USDT) is outside the base

12 · Variances, by cause

```
expected_out = converted_in × partner_rate_version.rate (as priced)
actual_out = what the partner booked (partner_calculates) or expected_out (npl_calculates)
variance = expected_out − actual_out (+ = NPL received less than priced)
client_net unchanged (write-once); pool = actual_out − client_net; the variance posts to
 VAR_CONVERSION partner_calculates differences and basis variance on a reroute
 VAR_CUTOFF rate stamped at lock vs rate used after the partner's cutoff (conversion_due_date)
 VAR_RATE_HONOUR an expired quote's rate honoured by override vs the day's rate
 VAR_FX_TIMING an obligation priced on one day and discharged on another (§14)
```

- partner_calculates: 3,000 USDT expected at 0.8910 = 2,673.00, booked at 0.8905 = 2,671.50; variance = 1.50 → VAR_CONVERSION; client net 2,646.00 unchanged
- cutoff timing: locked 0.8910, converted next day at 0.8890 → 6.00 → VAR_CUTOFF
- honoured expired rate: 0.8910 honoured, Ali at 0.8860 → 15.00 → VAR_RATE_HONOUR

13 · Earnings retained in the collection currency and offset between partners

```
retained_in = earnings ÷ partner_rate (this deal's margin kept unconverted, residual_in)
offset_to = from_amount ÷ to_partner_rate_of_day (EARNINGS_OFFSET: NPL's margin at the from-
partner, in USDT at the to-partner)
converted_in = amount_in − retained_in − offset_to
client_out = converted_in × partner_rate = client_net − from_amount already delivered at the
from-partner
neutral for the client: Σ client entitlement unchanged; NPL's EARN_GROSS for the deal is
recognised in the collection currency
```

- NPL's margin at Jeton (EUR) 2,025.00 ÷ Ali's rate of the day 0.8910 = 2,272.73 USDT taken from the client's unconverted balance at Ali
- Ali deal margin 900.00 EUR ÷ 0.8910 = 1,010.10 USDT retained (residual_in)
- Ali converts 100,000 − 2,272.73 − 1,010.10 = 96,717.17 USDT × 0.8910 = 86,175.00 EUR = client net 88,200.00 − 2,025.00 already delivered at Jeton

14 · Invoices paid in instalments and the forex true-up

```
reserved_i = obligation_discharged_i × DEAL_GROUP.invoice_rate (payout currency per invoice
currency at pricing: the stored base)
paid_i = obligation_discharged_i × SETTLEMENT_LINE.discharge_rate_i (same direction, at
payment)
true_up_i = reserved_i − paid_i (+ gain, − loss) → VAR_FX_TIMING against NPL's pool
INVOICE_BALANCE = vendor invoice: amount − Σ obligation_discharged over released lines not
returned; resale invoice: amount − Σ DEAL_GROUP.invoice_amount collected
```

- invoice_rate = 1.148369 USDT per EUR (0.8708 the other way); instalment 1: 3,000 EUR × 1.148369 = 3,445.11 USDT paid at the same rate it was reserved at; true-up 0
- instalment 2: discharge_rate 1.142505; paid 6,000 × 1.142505 = 6,855.03; reserved 6,000 × 1.148369 = 6,890.22; true-up = 35.19 USDT gain → VAR_FX_TIMING (a debit when the rate moves against NPL)
- margin line: 450 EUR × 1.142505 = 514.13; reserved 450 × 1.148369 = 516.77; true-up 2.64

15 · The reseller project (NPL-GR)

```
reseller_margin = resale_invoice − its vendor_invoice (INVOICE.vendor_invoice_id) (EUR; fixed
when the two invoices are posted)
target_out = resale_invoice ÷ pricing_rate (USDT to deliver)
amount_in = target_out × (1 + remittance_markup) ÷ market_rate (§6; cash-rounded if THB cash)
remittance_margin = earnings (§8) + Σ true_up_i (§14) − fees (NPL's example: 2 % − 1 % Ali −
0.1 % forex = 0.9 %…1.2 %)
NPL_PROFIT = Σ remittance_margin + Σ reseller_margin (per currency; one figure through
REPORTING_VALUE)
```

16 · Pools, squaring, loss splits, balance conversion, returns, network fees

```
squaring: pool_ccy < 0 after an absorbed fee → next conversion includes (−pool_ccy ÷
partner_rate) of NPL's own USDT
loss split: sender_share + npl_share + partner_share + client_share = LOSS_EVENT.amount ≤
amount verified or released; one LOSS_RECOVERY posting per share (T41a–c)
balance conv.: out = amount × partner_rate (no fee unless a FEE_DECISION applies §3); rounding
as the pair says
return leg: out = returned_amount × partner_rate; markup only by FEE_DECISION; the first
conversion's margin stays earned
bank fee: treatment waived (client bears) | carry_forward (SHORTFALL, topped up) | absorbed or
covered (NPL, out of the pool)
network fee: USDT; zero and unposted when none applies; absorbed by NPL or charged back to the
sender per project policy
```

- bounce fee 15.00 GBP absorbed with no GBP pool at Ali → pool −15.00; squared at the next conversion: 15.00 ÷ 0.7722 = 19.43 USDT of NPL's own converted alongside the client's
- loss 10,000.00 USDT → EXP_LOSS; split partner 4,000.00 + sender 2,000.00 + client 1,000.00 + NPL 3,000.00 = 10,000.00; EXP_LOSS after recoveries = 3,000.00

17 · Reporting values

```
REPORTING_VALUE(figure, ccy, as_of) = figure × latest market_rate(platform pair ccy →
reporting_currency), with the rate, its time and a stale flag
never posted, quoted or settled; the only control reading it is a ceiling stated in the
reporting currency
```

18 · Acceptance vectors A — the ledger's worked deals

Produced by the pricing function the Ledger Posting Design imports; the posting tables there show the same figures line by line. Rates: market 0.9000 EUR per USDT, Ali 0.8910; Evo sender 1.4 %, receiver 0.6 % (L3: 1.4 / 1.4; L6: 4 % / 0; L7: 2 % markup on the target, receiver 0). L10 to L13 are the cases added in the design review: a part collection, an eight-decimal asset and two deals on a tier boundary; figures to the cent, BTC amounts to eight decimals. L14 is the case added in round 2 of the review: a deal priced on the opening day that straddles the tier threshold, with the opening volume in the counter (§5a).

#	Case	amount_in	sender rate	gross out	sender part	receiver part	client net	delivered	partner cost	NPL earnings	payable	residual
L1	Deal 1 — Evo, 10,000 USDT → EUR at Ali (fee_in_rate), market 0.9000, Ali 0.8910, sender 1.4 % + receiver 0.6 % on market	10,000.00 USDT	0.887400	8,874.00	126.00	54.00	8,820.00	8,910.00	0.00	90.00	8,820.00	0.00
L2	Deal 3 — 10,000 USDT → EUR at Jeton (fee_on_market 1.25 %), same fee	10,000.00 USDT	0.887400	8,874.00	126.00	54.00	8,820.00	8,887.50	112.50	67.50	8,820.00	0.00
L3	Same-currency 5,000 EUR → EUR at rate 1, Evo EUR → EUR 2.8 % split 1.4 / 1.4	5,000.00 EUR	0.986000	4,930.00	70.00	70.00	4,860.00	5,000.00	0.00	140.00	4,860.00	0.00
L4	3,000 USDT → THB cash at Ali (market 32.60, Ali 32.27), round_up_n 1,000	3,000.00 USDT	32.143600	96,430.80	1,369.20	586.80	95,844.00	96,810.00	0.00	966.00	96,000.00	-156.00
L5	2,000 USDT → GBP at Ali (market 0.7800, Ali 0.7722)	2,000.00 USDT	0.769080	1,538.16	21.84	9.36	1,528.80	1,544.40	0.00	15.60	1,528.80	0.00
L6	Raeen sub-account: 10,000 USDT → EUR at Aquanow (fee_on_market 0.80 %), sender 4 % on market, receiver 0	10,000.00 USDT	0.864000	8,640.00	360.00	0.00	8,640.00	8,928.00	72.00	288.00	8,640.00	0.00
L7	NPL-GR: target 10,852.09 USDT (= 9,450 EUR at 0.8708) collected in THB cash at Ali, 2 % markup on the target, cash round-up 1,000; Ali 1 ÷ 32.83 vs market 1 ÷ 32.50	360,000.00 THB	0.030166	10,859.73	217.19	0.00	10,859.73	10,966.15	0.00	106.42	10,859.73	0.00
L8	Pattern N: 300,000 USDT → EUR at Jeton (1.25 %)	300,000.00 USDT	0.887400	266,220.00	3,780.00	1,620.00	264,600.00	266,625.00	3,375.00	2,025.00	264,600.00	0.00
L9	Pattern N: 100,000 USDT → EUR at Ali	100,000.00 USDT	0.887400	88,740.00	1,260.00	540.00	88,200.00	89,100.00	0.00	900.00	88,200.00	0.00

#	Case	amount_in	sender rate	gross out	sender part	receiver part	client net	delivered	partner cost	NPL earnings	payable	residual
L10	Part collection: deal 2, 5,000 USDT → EUR at Ali collected as 4,000 + 1,000 (T4 to T6); priced once on the whole amount, nothing posts on the parts	5,000.00 USDT	0.887400	4,437.00	63.00	27.00	4,410.00	4,455.00	0.00	45.00	4,410.00	0.00
L11	Eight-decimal asset: 0.12345678 BTC → EUR at Ali, market 60,000.00000000 EUR per BTC, Ali 59,400.00000000 (1 % inside), sender 1.4 % + receiver 0.6 %; rate_precision 8, amounts 2 dp	0.12345678 BTC	59,160.00000000	7,303.70	103.70	44.44	7,259.26	7,333.33	0.00	74.07	7,259.26	0.00
L12	Tier boundary, exactly reached: GDC, month-to-date USD 450,000.00, target_out USD 50,000.00 at rate 1; the deal ends the month-to-date at exactly 500,000.00, so the whole deal is tier 1 (2.5 %) and the next deal, of any size, is tier 2	51,250.00 USDT	0.975610	50,000.00	1,250.00	0.00	50,000.00	51,250.00	0.00	1,250.00	50,000.00	0.00
L13	Tier boundary, crossed: GDC, month-to-date USD 490,000.00, target_out USD 20,000.00; 10,000.00 at 2.5 % and 10,000.00 at 2.25 % (two DEAL_FEE_TIER portions, blended 2.375 %)	20,475.00 USDT	0.976801	20,000.00	475.00	0.00	20,000.00	20,475.00	0.00	475.00	20,000.00	0.00
L14	Opening volume in the counter: GDC opens on day 20 with USD 480,000.00 month-to-date (opening line), target_out USD 40,000.00 at rate 1; 20,000.00 at 2.5 % and 20,000.00 at 2.25 % (blended 2.375 %); counter afterwards 520,950.00	40,950.00 USDT	0.976801	40,000.00	950.00	0.00	40,000.00	40,950.00	0.00	950.00	40,000.00	0.00

L4's payable is the cash hand-over rounded up to 1,000 THB (residual negative, NPL's cost). L7's client net includes the 7.63 USDT cash-rounding surplus. Variances, offsets and the forex true-ups are in §12–§14 with their figures.

L11: principal 0.12345678 × 60,000 = 7,407.41; sender part 1.4 % = 103.70, receiver part 0.6 % = 44.44, client net 7,407.41 × 0.98 = 7,259.26 = gross out 7,303.70 − 44.44; delivered 0.12345678 × 59,400 = 7,333.33; earnings 74.07. L12 and L13: GDC converts at a rate of 1 and the partner's cost is inside it, so delivered = amount_in and NPL's earnings are the markup; the sender rate is printed as target ÷ amount_in. NPL supplied its own figures for these four cases on 21 October 2026; they are vectors B19 to B22 (§19) and all four match. L14: opening line 480,000.00 + target 40,000.00: portions 20,000.00 at 2.5 % and 20,000.00 at 2.25 %, amount_in 40,950.00, markup 950.00 (blended 2.375 %); counter afterwards 520,950.00 = 480,000.00 + 40,950.00. NPL confirmed these figures on 27 October 2026; they are vector B24 (§19) and it matches.

19 · Acceptance vectors B — NPL's own figures, recomputed

Each row takes the inputs NPL's fee sheet shows, applies this document's formulas, and compares with the figure the sheet prints. "matches" means to the cent, or to the unit where the sheet rounds. Rows are numbered B1 to B24 (19b). B2 is compared with NPL's corrected sheet of 21 October 2026; B18 to B23 were added on 21 October: the receiver side of the Evo selected-senders deal from the corrected sheet, and the five figures NPL supplied for the cases its sheet had no line for. B24 was added on 27 October 2026 in round 2 of the review: NPL's figure for the opening-volume case L14.

Id	Case	Formula applied	Computed	NPL's sheet	Result	Note
B1	Evo sender: 29,325 USDT at market 0.888977323, sender pays 1.40 %	fixed_in; sender rate = market × (1 − 1.40 %) = 0.876531640478	25,704	25,704	**matches**	NPL's sheet prints the sender rate 0.876531640478 and EUR 25,704 (rounded to the unit)
B2	Evo receiver: the same deal, receiver pays 0.60 %	fee parts on the principal: 29,325 × 0.888977323 × (1 − 1.40 % − 0.60 %)	25,548	25,548	**matches**	corrected sheet of 21 October: (29,325 − 2 %) × 0.888977323 = 25,548; the 8 October sheet read 25,550 (receiver part taken on the gross out)
B3	Evo selected senders: EUR 50,000 target, market 1 ÷ 0.8652 USDT per EUR, sender pays 1.00 %	target_out; amount_in = target × (1 + 1.00 %) × market in_per_out = 50,000 × 1.01 × 1.155802126676	58,368	58,368	**matches**	NPL prints the final rate 1.16736014794267 = market × 1.01
B4	BF-Raeen via Jeton: 200,000 USDT at 0.8610999, fee 1 %	fixed_in; 200,000 × 0.8610999 × (1 − 1 %)	170,498	170,498	**matches**	
B5	BF-Raeen via Ali: 300,000 USDT at 0.864189, fee 1 %	fixed_in; 300,000 × 0.864189 × (1 − 1 %)	256,664	256,664	**matches**	the printed rate is recomputed so the sender-facing fee reads 1 %
B6	Raeen sub-account: 50,000 USDT at 0.8650036, fee 4 %	fixed_in; 50,000 × 0.8650036 × (1 − 4 %)	41,520.17	41,520.17	**matches**	
B7	GDC: USD 84,482 target, fee 2.5 %, converted at a rate of 1 USDT per USD	target_out; 84,482 × (1 + 2.5 %) ÷ 1.0000	86,594	86,594	**matches**	the sheet's figure implies a rate of exactly 1
B8	GDC above USD 500,000 a month: tier 2.25 %	target_out; 84,482 × (1 + 2.25 %)	86,383	86,383	**matches**	FEE_TIER chosen from PROJECT_MONTH_VOLUME
B9	27 Group (reseller): THB 17,000,000 at 0.0259463782412186 EUR per THB, fee 2 %	fixed_in; sender rate = 0.0259463782412186 × 0.98 = 0.02542745; 17,000,000 × 0.02542745	432,266.66	432,266.66	**matches**	NPL rounds the EUR to the unit: 432,267
B10	27 Group: the same THB at Ali, agent rate 1 ÷ 33.83 USDT per THB	what the partner delivers: 17,000,000 ÷ 33.83	502,513	502,513	**matches**	agent fee 0.64 % inside the rate against market 1 ÷ 33.613

Id	Case	Formula applied	Computed	NPL's sheet	Result	Note
B11	27 Group balance clearance: EUR 329,408 target, market 38.13570206 THB per EUR, fee 2 %	target_out; 329,408 × 1.02 × 38.13570206	12,813,449.45	12,813,449.45	**matches**	then cash-rounded by NPL to THB 12,813,000 (nearest 1,000; the pair's cash_rounding)
B12	Bala (reseller): EUR 2,220 target, market 1 ÷ 0.8716, fee 2 %	target_out; 2,220 × 1.02 × 1.147315282240	2,597.98	2,597.98	**matches**	then rounded up to USDT 2,600 (USDT paid by reseller customers rounded up)
B13	66 Group (reseller): EUR 239,149 target, market 1 ÷ 0.8796, fee 1.5 %	target_out; 239,149 × 1.015 × 1.136880400182	275,962.07	275,962.07	**matches**	rounded to USDT 275,962
B14	TripleSeven/GZ (reseller): EUR 4,097 target, market 1 ÷ 0.8598, fee 1.5 %	target_out; 4,097 × 1.015 × 1.163061177018	4,837	4,837	**matches**	
B15	Alpha Plus (reseller): EUR 2,939 target, market 1 ÷ 0.8712, fee 1.5 %	target_out; 2,939 × 1.015 × 1.147842056933	3,424.11	3,424.11	**matches**	rounded to USDT 3,424
B16	Fair (reseller): EUR 1,079 target, market 1 ÷ 0.8642, fee 0.75 %	target_out; 1,079 × 1.0075 × 1.157139551030	1,257.92	1,257.92	**matches**	rounded to USDT 1,258
B17	rate direction: Ali quotes 1.1738 (USDT per EUR) on a USDT → EUR pair	normalised rate = 1 ÷ 1.1738 at the pair's precision (6)	0.851934	0.851934	**matches**	a direction slip freezes the pair instead of pricing a deal
B18	Evo selected senders, receiver side: EUR 50,000 target, receiver pays 0.60 %	receiver's part on the target: 50,000 × (1 – 0.60 %)	49,700	49,700	**matches**	the receiver line of B3; corrected sheet of 21 October
B19	Part collection: 5,000 USDT → EUR at Ali, received as 4,000 + 1,000, priced once on the whole amount	fixed_in; gross out 5,000 × 0.9000 × (1 – 1.4 %); client net 5,000 × 0.9000 × (1 – 2 %)	4,437.00 / 4,410.00	4,437 / 4,410	**matches**	NPL's figure of 21 October, sender side / receiver side; nothing posts on the parts (L10)
B20	Eight-decimal asset: 0.12345678 BTC → EUR at Ali, market 60,000, Ali 59,400	fixed_in; gross out 0.12345678 × 60,000 × (1 – 1.4 %); client net 0.12345678 × 60,000 × (1 – 2 %)	7,303.70 / 7,259.26	7,304 / 7,259	**matches**	NPL's figure of 21 October, rounded to the unit (L11)
B21	Tier reached exactly: GDC, month-to-date USD 450,000, USD 50,000 target out	target_out; 50,000 × (1 + 2.5 %), tier 1 on the whole deal	51,250	51,250	**matches**	NPL's figure of 21 October; the next deal of any size is tier 2 (L12)
B22	Tier crossed inside one deal: GDC, month-to-date USD 490,000, USD 20,000 target out	target_out; 10,000 × (1 + 2.5 %) + 10,000 × (1 + 2.25 %)	20,475	20,475	**matches**	NPL's figure of 21 October, the two-portion split confirmed (L13)
B23	Same-currency: 5,000 EUR → EUR at rate 1, Evo 2.8 % split 1.4 / 1.4	fixed_in at rate 1; client net 5,000 × (1 – 2.8 %)	4,860	4,860	**matches**	NPL's figure of 21 October (L3)
B24	Opening tier volume: GDC opens on day 20 at USD 480,000 month-to-date, USD 40,000 target out	target_out; 20,000 × (1 + 2.5 %) + 20,000 × (1 + 2.25 %); counter afterwards 480,000 + 40,950	40,950	40,950	**matches**	NPL's figure of 27 October 2026: amount in 40,950.00, markup 950.00, counter 520,950.00 (L14)

24 of 24 match.

19a · Machine-readable vectors and the regression gate

Vector sets A (L1 to L14) and B (B1 to B24, numbered as in the Reading Companion's identifier appendix) are published with this document as one machine-readable file (`vectors-v5.9.csv`: one row per vector with every input and every expected figure), which is the file the automated tests read. Every release runs the full vector set together with the ledger fixtures (the 61 worked transactions), the state-machine tests and the visibility tests; a failing exact test blocks the release. The gate is stated in the Test Strategy & Coverage document and continues under the support bundle after go-live; the first automated runs are in Phase 1 (M4).

19b · Identifiers of the vector B rows

The rows of Section 19 are numbered B1 to B24 in the order printed, so that the test suite, the Reading Companion's identifier register and NPL's sheet can refer to one row by one name. The numbers are frozen: a row is never renumbered and a number is never reused; a new case takes the next number (B25).

19c · Which cases the two vector sets cover

A vector is one worked deal: its inputs (amount, rates, fee parts) and the figure the system must produce. Set A (L1 to L14, Section 18 and 18a) are examples New XP wrote and worked through the pricing formula by hand; they prove the formula does what this specification says, not that NPL agrees with the result. Set B (B1 to B24, Section 19) are NPL's own numbers: each row takes a line from NPL's fee sheet of 8 October (corrected 21 October) or a figure NPL supplied on 21 October or 27 October for the cases the sheet had no line for, recomputes it with the formula and shows whether the two agree. A matching B row means the system reproduces what NPL charges today, which is the acceptance test NPL cares about; a matching A row means the system matches our reading of the rules.

The table below puts the two sets side by side, by case, so that one page answers what is covered by NPL's own figure and what is not.

Case	Vector A	Vector B	Status
Fixed amount in, partner fee inside the rate, sender and receiver parts	L1, L5, L9	B1, B2, B5	covered (B2 on the corrected sheet of 21 October, 19a)
Fixed amount in, partner fee on the market price (Jeton, Aquanow)	L2, L8	B4	covered
Sub-account with a markup share and a partner rebate	L6	B6	covered
Target amount out, markup on the target; sender and receiver sides	L7	B3, B18, B12 to B16	covered
Cash delivery rounded up to the note	L4	B11	covered
Agent rate: the partner's fee inside the rate it quotes	L7	B10	covered
Reseller deal priced from a fixed amount in	—	B9	covered by NPL's figure; no hand-worked example needed
Monthly tier: tier 1 and tier 2 (GDC)	L12, L13	B7, B8	covered
Rate direction: a pair quoted the other way round	—	B17	covered by NPL's figure

Case	Vector A	Vector B	Status
Same-currency deal at rate 1 (Evo EUR → EUR)	L3	B23	covered
Deal collected in parts, priced once on the whole	L10	B19	covered
Asset quoted to eight decimals (BTC)	L11	B20	covered
Monthly tier reached exactly on the deal	L12	B21	covered
Monthly tier crossed inside one deal, two portions	L13	B22	covered (NPL confirms the two-portion split)
Opening tier volume: a deal straddling the threshold on the opening day (§5a)	L14	B24	covered: NPL's figure of 27 October 2026 matches

Every case now has a B row, L14 included: NPL confirmed its figures on 27 October 2026 and they are B24. NPL supplied the figures for the five cases its sheet had no line for on 21 October 2026, each worked on its own sheet convention (sender side and receiver side, rounded to the unit); all five agree with the formula and are frozen as B19 to B23 under the rule of 19b. A new case takes B25.

20 · Acceptance

1. The engine reproduces every vector of §18 and §19 exactly (to the cent, or to the unit where the sheet rounds), and every posting amount of the Ledger Posting Design.
2. Fee parts are computed on the principal (§3) and never compounded; a target-out quote applies the fee as a markup on the target (§6).
3. Every rounding step uses the pair's configured option and records the signed residual with its owner (§9).
4. Variances are posted to the account of their cause (§12); the forex true-up of each instalment is posted per line (§14); an earnings offset leaves every client entitlement unchanged (§13).
5. NPL_PROFIT returns remittance earnings, reseller earnings and their sum, per currency, and the same figures through REPORTING_VALUE (§15, §17).

Vectors A are worked examples; vectors B are NPL's figures from the fee table of 8 October 2026.